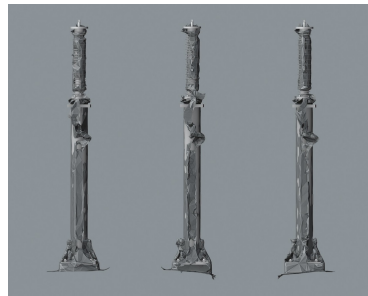


# Why the Reconstruction Loop Broadens: Three Structural Transitions

Each row shows how a new dependency changes the engineering decision and therefore the operator choice.

A · Fit →  
reusable  
scope

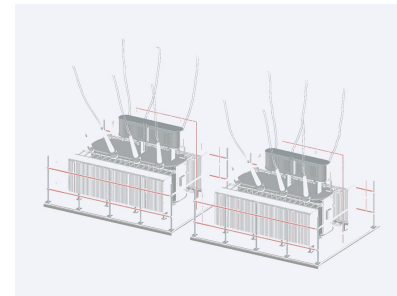
## B01: local fit



define a local comparison domain, then fit pose and size

repetition →  
invariance

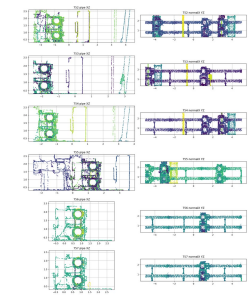
## B08: shared equipment



two installations reuse one shared MASTER

reuse boundary becomes testable

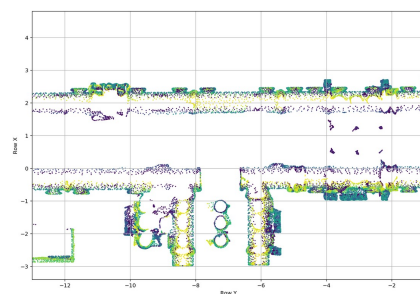
## B15: revise reuse boundary



finite site geometry moves outside the shared asset

B · Object →  
connected  
representation

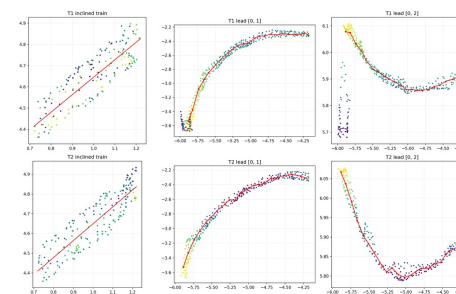
## B20: ports and routes



existing equipment ends become explicit ports

relationships appear

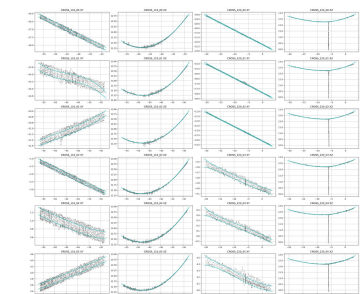
## B23: change representation



the connection is reformulated from straight to curved

endpoint/topology constraints

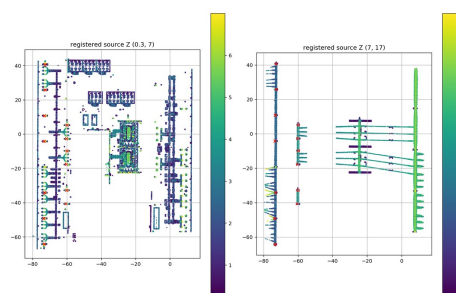
## B29: relational constraints



continuity and endpoint constraints scale across subsystems

C · Addition →  
state-driven  
closure

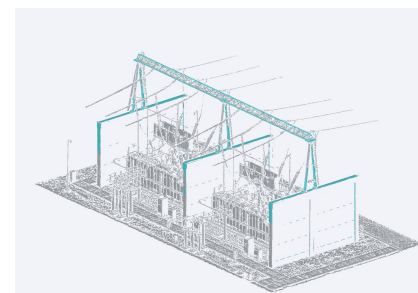
## B25: unexplained geometry



large unexplained structures remain in the scene

coverage → task selection

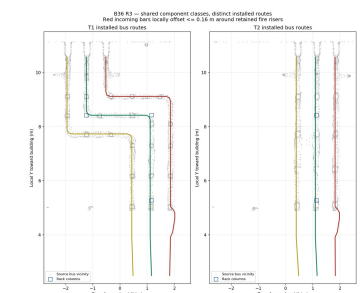
## B26: reconstruct selected gaps



coverage selects which missing structures to reconstruct next

fit the inherited world

## B36: state-constrained insertion



new geometry must fit inherited endpoints and occupied space

Descriptive evidence from the B01–B36 record. The figure explains how the reconstruction problem changes; it does not propose a universal complexity taxonomy.